`GeoModels` Tutorial: simulation, estimation and prediction of spatial binomial data using binomial random fields.

Moreno Bevilacqua
Christian Caamaño-Carrillo

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Introduction

In this tutorial we show how to analyze spatial discrete data using random fields with binomial marginal distribution using the R package `GeoModels` [Bevilacqua et al., 2026]. We first load the R libraries needed in this tutorial.

```
rm(list=ls());
require(GeoModels);
require(fields);
# The model is specified explicitly in GeoSim and GeoFit.
```

Binomial random fields

We first define a Bernoulli random field. A simple approach for defining a Bernoulli random field $B = \{B(\boldsymbol{s}), \boldsymbol{s} \in A\}$ is by thresholding a zero mean and unit variance Gaussian random field $G = \{G(\boldsymbol{s}), \boldsymbol{s} \in A\}$ with correlation function $\mathrm{cor}\{G(\boldsymbol{s}), G(\boldsymbol{s}+\boldsymbol{h})\} = \rho(\boldsymbol{h})$, namely,

$$B(\boldsymbol{s}) = \begin{cases} 1, & G(\boldsymbol{s}) < \mu(\boldsymbol{s}), \\ 0, & \text{otherwise}, \end{cases}$$

where $\mu(\boldsymbol{s})$ is a nonrandom function. With this definition the marginal probability of success is

$$p(\boldsymbol{s}) := \Pr\{B(\boldsymbol{s}) = 1\} = \Phi\{\mu(\boldsymbol{s})\}, \tag{1}$$

where Φ is the univariate standard Gaussian cumulative distribution function.

The main idea to obtain a random field with binomial marginal distribution is to sum, for each $\boldsymbol{s}$, $n(\boldsymbol{s})$ independent copies of B; that is, we define a random field $Y = \{Y_{n(\boldsymbol{s})}(\boldsymbol{s}), \boldsymbol{s} \in A\}$ as

$$Y_{n(\boldsymbol{s})}(\boldsymbol{s}) := \sum_{i=1}^{n(\boldsymbol{s})} B_i(\boldsymbol{s}), \tag{2}$$

where $n(\boldsymbol{s})$ is a positive integer. Then, at each site $\boldsymbol{s}$, the binomial random field represents the number of successes in $n(\boldsymbol{s})$ independent trials. Its marginal distribution is binomial:

$$\Pr\{Y_{n(\boldsymbol{s})}(\boldsymbol{s}) = u\} = \binom{n(\boldsymbol{s})}{u} p(\boldsymbol{s})^u \{1 - p(\boldsymbol{s})\}^{n(\boldsymbol{s})-u}, \quad u = 0, 1, \ldots, n(\boldsymbol{s}).$$

The mean and variance are $\mathrm{E}\{Y_{n(\boldsymbol{s})}(\boldsymbol{s})\} = n(\boldsymbol{s})p(\boldsymbol{s})$ and $\mathrm{Var}\{Y_{n(\boldsymbol{s})}(\boldsymbol{s})\} = n(\boldsymbol{s})p(\boldsymbol{s})\{1-p(\boldsymbol{s})\}$. Moreover, the *covariance* function is given by

$$\mathrm{Cov}\{Y_{n(\boldsymbol{s}_i)}(\boldsymbol{s}_i), Y_{n(\boldsymbol{s}_j)}(\boldsymbol{s}_j)\} = \min\{n(\boldsymbol{s}_i), n(\boldsymbol{s}_j)\}\{p_{11}(\boldsymbol{s}_i, \boldsymbol{s}_j) - p(\boldsymbol{s}_i)p(\boldsymbol{s}_j)\}, \tag{3}$$

where p_{11} is the probability of joint success at locations $\boldsymbol{s}_i, \boldsymbol{s}_j$ in the Bernoulli field, namely,

$$p_{11}(\boldsymbol{s}_i, \boldsymbol{s}_j) = \Pr\{B(\boldsymbol{s}_i) = 1, B(\boldsymbol{s}_j) = 1\} = \Phi_2\{\mu(\boldsymbol{s}_i), \mu(\boldsymbol{s}_j); \rho(\boldsymbol{s}_i - \boldsymbol{s}_j)\},$$

and $\Phi_2(\cdot, \cdot; \rho)$ is the bivariate standard Gaussian CDF with correlation ρ. The corresponding correlation is obtained by dividing (3) by the product of the marginal standard deviations. When $p(\boldsymbol{s})$ and $n(\boldsymbol{s})$ are constant, Y is stationary.

Stationary Binomial random fields

We first set the spatial coordinates:

```
set.seed(269);
N=1000;
coords=cbind(runif(N),runif(N));
```

Let us assume $\mu(\boldsymbol{s}) = \mu$ and $n(\boldsymbol{s}) = n$. Then the probability of success $p(\boldsymbol{s}) = p$ is constant and the binomial random field has constant mean and variance. We specify the probability through μ, the number of trials n, and a parametric correlation model for the underlying Gaussian random field.

For the correlation function $\rho(\boldsymbol{h})$ of the parent Gaussian field we consider an isotropic Matérn model [Matérn, 1986]:

$$\rho_{\alpha,\gamma}(\boldsymbol{h}) = \frac{2^{1-\gamma}}{\Gamma(\gamma)} \left(\frac{\|\boldsymbol{h}\|}{\alpha} \right)^{\gamma} K_{\gamma} \left(\frac{\|\boldsymbol{h}\|}{\alpha} \right), \quad \|\boldsymbol{h}\| \geq 0, \tag{4}$$

where K_γ is the modified Bessel function of the second kind of order γ, $\gamma > 0$ the smoothness parameter and $\alpha > 0$ the scale parameter. We use:

```
corrmodel="Matern";
scale=0.25/3;
smooth=0.5;
nugget=0;
```

We fix the number of trials $n = 10$ and the probit-scale parameter $\mu = 0.5$, which gives $p = \Phi(0.5) \simeq 0.691$:

```
n=10;
mean=0.5;
param=list(nugget=nugget,mean=mean,scale=scale,smooth=smooth,sill=1);
data_s <- GeoSim(coordx=coords,corrmodel=corrmodel,
                          param=param,model="Binomial",n=n)$data;
```

The empirical mean and variance in the original illustration were close to the corresponding theoretical values (the numbers printed below are historical):

```
mean(data_s); var(data_s)
# 6.985; 2.229004
p=pnorm(mean)
n*p; n*p*(1-p)
# 6.914625; 2.133421
```

The following figure shows the histogram, the colored map and the empirical semivariogram for the stationary binomial field. The plots are retained from the original tutorial.

```
par(mfrow=c(1,3));
plot(table(data_s),ylab="Frequency");
quilt.plot(coords,data_s,nlevel=n+1,zlim=c(0,n));
vario_s=GeoModels::GeoVariogram(coordx=coords,data=data_s,maxdist=0.6);
plot(vario_s,xlab="h",ylab=expression(gamma(h)),
    ylim=c(0,max(vario_s$variograms)),pch=20,
    main="Semi-variogram");
```

Non-stationary Binomial random fields

A nonstationary binomial random field can be specified by assuming that $\mu(\boldsymbol{s})$ and/or $n(\boldsymbol{s})$ are not constant. We take a linear specification $\mu(\boldsymbol{s}) = X(\boldsymbol{s})^{\mathsf{T}}\boldsymbol{\beta}$, with $k = 2$ covariates, and use the Matérn correlation model in (4) with the same parameters. We set the regression parameters and the regression matrix:

```
mean=0.3;
mean1=-0.25;
```

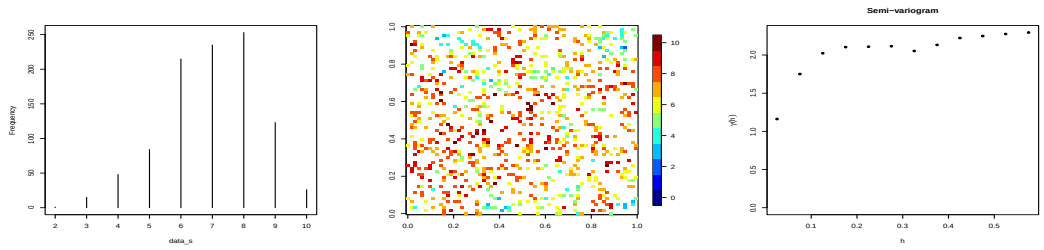


Figure 1: Histograms and coloured maps of a realization from a stationary binomial random field. On the right the associated empirical semivariogram estimation.

```
set.seed(29);
a0=rep(1,N); a1=runif(N);
X=cbind(a0,a1);
```

For the location-specific numbers of trials, we use

```
n=sample(10:20,nrow(coords),replace=TRUE);
head(n);
# 15 19 18 10 18 19
```

The nonstationary binomial field is simulated with

```
param=list(nugget=nugget,mean=mean,mean1=mean1,
          scale=scale,smooth=smooth,sill=1);
data_ns <- GeoSim(coordx=coords,corrmodel=corrmodel,
                          param=param,n=n,X=X,model="Binomial")$data;
```

Estimation of binomial random fields

Let y_j denote an observed realization of $Y_{n_j}(\boldsymbol{s}_j)$; for notation simplicity set $n_j = n(\boldsymbol{s}_j)$, $p_j = p(\boldsymbol{s}_j)$ and $\mu_j = \mu(\boldsymbol{s}_j)$. Given observations $\boldsymbol{y}_n = (y_1, \ldots, y_l)^\top$ at sites $\boldsymbol{s}_1, \ldots, \boldsymbol{s}_l$, regression and correlation parameters can be estimated using composite likelihood based on pairs. The *conditional* pairwise log-likelihood is

$$\mathrm{cpl}(\boldsymbol{\theta}) = \sum_{i=1}^{l} \sum_{\substack{j=1 \\ j \neq i}}^{l} c_{ij} \left[\log \Pr\{Y_{n_i}(\boldsymbol{s}_i) = y_i, Y_{n_j}(\boldsymbol{s}_j) = y_j\} - \log \Pr\{Y_{n_j}(\boldsymbol{s}_j) = y_j\} \right]. \qquad (5)$$

The bivariate probability mass function (see Marshall and Olkin [1985]), denoted by

$$f_{ij}(y_i, y_j) = \Pr\{Y_{n_i}(\boldsymbol{s}_i) = y_i, Y_{n_j}(\boldsymbol{s}_j) = y_j\},$$

can be written as

$$f_{ij}(y_i, y_j) = \begin{cases} \sum_{k=0}^{n_i - n_j} \binom{n_i - n_j}{k} p_i^k (1-p_i)^{n_i - n_j - k}\, b(y_i - k, y_j; p_i, p_j; n_j), & n_i > n_j, \\ b(y_i, y_j; p_i, p_j; n_i), & n_i = n_j, \\ \sum_{k=0}^{n_j - n_i} \binom{n_j - n_i}{k} p_j^k (1-p_j)^{n_j - n_i - k}\, b(y_j - k, y_i; p_j, p_i; n_i), & n_j > n_i, \end{cases}$$

where, writing $p_{11} = \Phi_2(\mu_i, \mu_j; \rho(\boldsymbol{s}_i - \boldsymbol{s}_j))$,

$$b(x_1, x_2; v_1, v_2; N) = \sum_{a=\max(0, x_1+x_2-N)}^{\min(x_1,x_2)} \frac{N!}{a!(x_1-a)!(x_2-a)!(N-x_1-x_2+a)!} \times p_{11}^a (v_1 - p_{11})^{x_1-a} (v_2 - p_{11})^{x_2-a} (1 + p_{11} - v_1 - v_2)^{N-x_1-x_2+a}.$$

The weights are chosen from nearest-neighbour relations:

$$c_{ij}(k) = \begin{cases} 1, & \boldsymbol{s}_i \in E_k(\boldsymbol{s}_j), \\ 0, & \text{otherwise}, \end{cases} \tag{6}$$

where $E_k(\boldsymbol{s}_j)$ denotes the k nearest neighbours of $\boldsymbol{s}_j$. The directed weights in (6) need not be symmetric. The parameter vector is $\boldsymbol{\theta} = (\boldsymbol{\beta}^\top, \alpha)^\top$ when smoothness and nugget are fixed. The conditional pairwise likelihood estimator maximizes (5).

Optimization is performed using `nlminb` with lower and upper bounds:

```
fixed2=list(sill=1,nugget=0,smooth=smooth);
start2=list(mean=mean,mean1=mean1,scale=scale);
lower=list(mean=-5,mean1=-5,scale=1e-6);
upper=list(mean=5,mean1=5,scale=10);
fit1_ns <- GeoFit(data=data_ns,coordx=coords,
           corrmodel=corrmodel,likelihood="Conditional",
           type="Pairwise",n=n,X=X,neighb=4,
           sensitivity=TRUE,optimizer="nlminb",
           lower=lower,upper=upper,
           start=start2,fixed=fixed2,model="Binomial");
```

Here `neighb=4` sets $k = 4$ in the weight function (6). The estimated parameters below are from the original 2025 example. They were not recalculated with the code above:

```
# Model Binomial; Conditional Pairwise; Matern; nlminb
# convergence: Successful; composite log-likelihood: -7668.58
# mean mean1 scale
# 0.31756 -0.23810 0.07267
```

Standard errors can be estimated using the parametric bootstrap via `GeoVarestbootstrap`; this simulation-based procedure can be time-consuming:

```
fit1_ns_sd <- GeoVarestbootstrap(fit1_ns,K=100,
                 optimizer="nlminb",lower=lower,upper=upper);
fit1_ns_sd$stderr
# mean: 0.044853355; mean1: 0.028044457; scale: 0.009831758
```

Prediction of binomial random fields

For a prediction site $\boldsymbol{s}_0$ with covariates $X(\boldsymbol{s}_0)$ and number of trials n_0, the optimal linear predictor of the binomial random field is

$$\widehat{Y}_{n_0}(\boldsymbol{s}_0) = \mathrm{E}\{Y_{n_0}(\boldsymbol{s}_0)\} + \boldsymbol{c}^{\mathsf{T}}\boldsymbol{\Sigma}^{-1}(\boldsymbol{y}_n - \boldsymbol{m}), \tag{7}$$

where $\boldsymbol{m} = (\mathrm{E}\{Y_{n_1}(\boldsymbol{s}_1)\}, \ldots, \mathrm{E}\{Y_{n_l}(\boldsymbol{s}_l)\})^{\mathsf{T}}$, $\mathrm{E}\{Y_{n_i}(\boldsymbol{s}_i)\} = n_i p_i$, and

$$\boldsymbol{c} = \big(\mathrm{Cov}\{Y_{n_0}(\boldsymbol{s}_0), Y_{n_i}(\boldsymbol{s}_i)\}\big)_{i=1}^{l}, \quad \boldsymbol{\Sigma} = \big[\mathrm{Cov}\{Y_{n_i}(\boldsymbol{s}_i), Y_{n_j}(\boldsymbol{s}_j)\}\big]_{i,j=1}^{l}.$$

The covariance vector and matrix are obtained using (3). The associated mean squared error, for fixed model parameters, is

$$\mathrm{MSE}\{\widehat{Y}_{n_0}(\boldsymbol{s}_0)\} = \mathrm{Var}\{Y_{n_0}(\boldsymbol{s}_0)\} - \boldsymbol{c}^{\mathsf{T}}\boldsymbol{\Sigma}^{-1}\boldsymbol{c}. \tag{8}$$

In practice both (7) and (8) are evaluated using the fitted parameters. We consider a regular grid of prediction locations and their associated covariates:

```
xx=seq(0,1,0.02);
loc_to_pred=as.matrix(expand.grid(xx,xx));
NN=nrow(loc_to_pred);
a0=rep(1,NN); a1=runif(NN);
Xloc=cbind(a0,a1);
nloc=sample(10:20,nrow(loc_to_pred),replace=TRUE);
```

Then the optimal linear prediction and the plug-in MSE can be computed using

```
param_est=as.list(c(fit1_ns$param,fixed2));
pr <- GeoKrig(data=data_ns,coordx=coords,
          loc=loc_to_pred,X=X,Xloc=Xloc,n=n,nloc=nloc,
          corrmodel=corrmodel,model="Binomial",
          mse=TRUE,param=param_est);
```

Finally, Figure 2 shows the original observed-data map, linear-kriging prediction and mean squared error, using:

```
par(mfrow=c(1,3));
nn=max(n);
quilt.plot(coords,data_ns,nlevel=nn+1,zlim=c(0,nn));
```

```
map=matrix(pr$pred,ncol=length(xx));
image.plot(xx,xx,map,xlab="",zlim=c(0,nn),ylab="",main="Kriging");
map_mse=matrix(pr$mse,ncol=length(xx));
image.plot(xx,xx,map_mse,xlab="",ylab="",main="MSE");
```

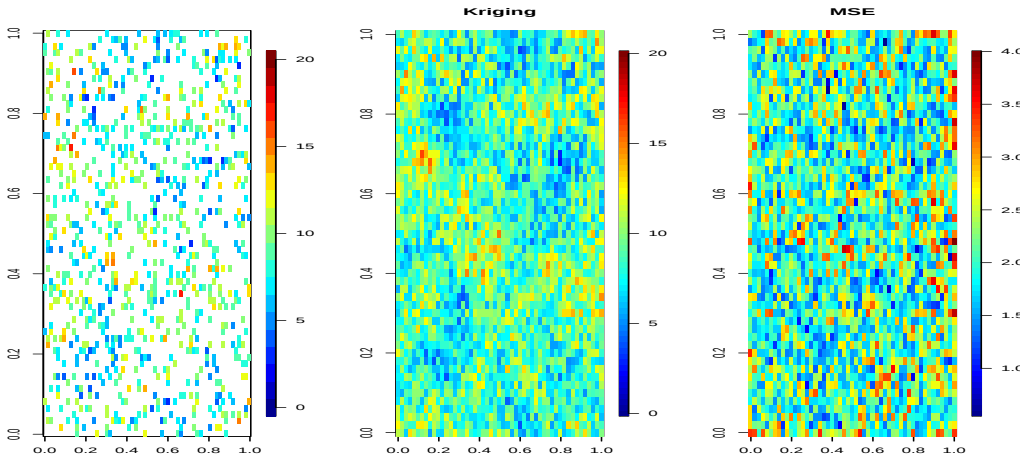


Figure 2: From left to right: observed spatial data, associated kriging map and mean square error map for a nonstationary binomial random field.

References

Moreno Bevilacqua, Christian Caamaño-Carrillo, Francisco Cuevas-Pacheco, and Víctor Morales-Oñate. *GeoModels: Procedures for Gaussian and Non Gaussian Geostatistical (Large) Data Analysis*, 2026. URL `https://CRAN.R-project.org/package=GeoModels`. R package version 2.3.0. DOI: 10.32614/CRAN.package.GeoModels.

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Bertil Matérn. *Spatial Variation: Stochastic Models and Their Application to Some Problems in Forest Surveys and Other Sampling Investigations*. Springer, 2 edition, 1986.